Data Profiling & Cleaning Report

Week 3 – Individual – Data Profiling
Dataset: tableau_prep_cleaning_exercise.csv

1. Dataset Import

The dataset was imported into Tableau Prep Builder using the Text File connector. The source file was located at data/week3/tableau_prep_cleaning_exercise.csv. A Clean Step was added to the flow to enable profiling and transformation.

2. Dataset Profiling

2.1 Basic Dataset Overview

Property	Value
Total rows (raw)	100
Total columns	18
Duplicate rows	5
Total missing values	76

2.2 Missing Values

The following columns contained null values identified during profiling:

Column	Missing Values	% of Rows
Satisfaction_Score	15	15%
Payment_Method	14	14%
Customer_Segment	12	12%
Email	10	10%
Notes	10	10%
Age	7	7%
Order_Date	5	5%
City	3	3%

2.3 Data Types

Unit_Price was stored as text (Abc) instead of a numeric type due to currency symbols and comma-decimal separators. Order_Date and Ship_Date were stored as text with mixed formats. All other columns had appropriate data types.

2.4 Value Distributions & Invalid Patterns

- **Age:** One value in the range 130-140 was detected — impossible for a real person.
- **Unit_Price:** Some values contained a Euro symbol (e.g. €215.16) and comma-decimal separators (e.g. 9999,99).
- **Order_Date:** Five or more different date formats were found: DD.MM.YYYY, DD/MM/YYYY, YYYY-MM-DD, YYYY/MM/DD, MM-DD-YYYY.

2.5 Duplicate Detection

Order_ID showed 95 unique values out of 100 rows, indicating 5 duplicate records. These were identified using a calculated field with PARTITION logic on Order_ID.

2.6 Cardinality of Key Columns

Column	Unique Values	Interpretation
Order_ID	95 (after dedup)	Primary identifier — nearly unique per row
Customer_ID	46	Foreign key — one customer can have many orders
Product_Category	10 (5 after cleaning)	Low cardinality — inflated by case noise
Customer_Segment	6 (3 after cleaning)	Low cardinality — inflated by case noise
Payment_Method	7 (5 after cleaning)	Low cardinality — inflated by case noise
Country	15	Medium cardinality — geographic dimension
Satisfaction_Score	5	Discrete scale 1-5

2.7 Inconsistent Text Formats

Three categorical columns had case inconsistencies where the same value appeared in multiple capitalisation formats:

- **Customer_Segment:** consumer / Consumer / CORPORATE / Corporate / home office / Home Office
- **Payment_Method:** bank transfer / Bank Transfer / credit card / Credit Card / PAYPAL / PayPal
- **Product_Category:** electronics / Electronics / ELECTRONICS / home / Home / furniture / Furniture etc.

3. Column Analysis

3.1 Columns with Many Unique Values

- Order_ID — 95 unique values (identifier)
- Unit_Price — 93 unique values (continuous numeric)
- Order_Date — 82 unique values (date)
- Sales — 92 unique values (continuous numeric)

3.2 Identifier Columns

- **Order_ID:** Best identifier for a single transaction — unique per order.
- **Customer_ID:** Identifies a customer across multiple orders — not unique per row.

3.3 Validity of Key Columns

Column	Valid?	Issue Found
Email	Yes	Format valid — contained nulls only
Order_Date	No	5+ different date formats mixed together
Product_Category	No	Case inconsistencies (fixed)
Customer_Segment	No	Case inconsistencies (fixed)
Age	No	One value of 130+ (invalid)
Unit_Price	No	Currency symbols and comma decimals
Sales	Yes	Values consistent after Unit_Price fix

4. Structure Analysis

4.1 Record Identification

- **Order_ID** can uniquely identify records after duplicate removal (95 unique values = 95 rows).
- **Customer_ID** cannot identify records alone 46 unique customers across 95 orders means customers repeat.

4.2 Calculated Field Consistency

The Sales column was checked against the formula: Quantity x Unit_Price x (1 - Discount). After cleaning Unit_Price (removing symbols and correcting format), values were consistent. No significant mismatches were found in the cleaned data.

5. Cleaning Rules Defined

Issue	Rule Applied	Rationale
Duplicate rows	Keep first occurrence, remove rest	Order data from original record is most reliable
Case inconsistencies	Group and Replace + Title Case	Standardise to single canonical value per category
Invalid Age (130+)	Exclude value (set to null)	Biologically impossible — treat as data entry error
Unit_Price symbols	Strip €, replace comma with period, convert to Float	Required for numeric calculations
Missing Customer_Segment	Fill with 'Unknown'	Preserve row — order data still valid
Missing Payment_Method	Fill with 'Unknown'	Preserve row — order data still valid
Missing City	Fill with 'Unknown'	Preserve row — order data still valid
Missing Age	Fill with mean (48)	Numeric imputation — mean appropriate for age
Missing Satisfaction_Score	Fill with mode (3)	Discrete scale — mode more appropriate than mean
Mixed date formats	DATEPARSE with conditional logic	Detect separator and year position to parse correctly

6. Cleaning & Transformation

6.1 Steps Performed in Tableau Prep

- **Removed duplicates:** Calculated field using PARTITION on Order_ID, filtered to keep Unique rows only. 5 rows removed.
- **Fixed case inconsistencies:** Group and Replace applied to Customer_Segment (3 values replaced), Payment_Method (3 values replaced), Product_Category (6 values replaced). Title Case applied to Customer_Segment and Product_Category.
- **Invalid Age excluded:** Filter applied to exclude Age values in the 130-140 range.
- **Unit_Price cleaned:** Calculated field using the formula FLOAT(REPLACE(REPLACE([Unit_Price], '€', ''), ',', '.')) to strip currency symbols and fix decimal separators. Original field removed and replaced.
- **Missing values filled:** Age nulls filled with 48 (mean). Satisfaction_Score nulls filled with 3 (mode). Customer_Segment, Payment_Method, City nulls filled with 'Unknown'.
- **Date standardisation:** Calculated field created using DATEPARSE with conditional logic to detect DD.MM.YYYY, DD/MM/YYYY, YYYY-MM-DD, YYYY/MM/DD, MM-DD-YYYY formats. Old Order_Date field removed and replaced with cleaned version.
- **Customer_Name standardised:** Group and Replace reduced 18 name variants to 12 (e.g. 'JOHN DOE' and 'john doe' merged). Title Case applied.
- **Country standardised:** Group and Replace reduced 10 country variants to 5 (e.g. 'FRANCE', 'france', 'France' merged). Title Case applied.
- **City standardised:** Group and Replace fixed 2 city inconsistencies. Title Case applied.

6.2 Calculated Fields Created

Field	Formula Logic	Purpose
Age_Group	IF Age <= 25 THEN 'Young' ELSEIF Age <= 40 THEN 'Adult' ELSEIF Age <= 60 THEN 'Middle Aged' ELSE 'Senior'	Bin ages into categories for analysis
Sales_Tier	IF Sales <= 100 THEN 'Low' ELSEIF Sales <= 500 THEN 'Medium' ELSEIF Sales <= 2000 THEN 'High' ELSE 'Premium'	Bin sales values for segmentation
Unit_Price_Scaled	(Unit_Price - 9.73) / (9999.99 - 9.73)	Min-max normalisation to 0-1 range
Order_Date_Clean	DATEPARSE with CONTAINS logic for separator detection	Standardise all date formats to Date type

7. Validation

Check	Before Cleaning	After Cleaning	Status
Row count	100	95	5 duplicates removed
Duplicate Order_IDs	5	0	Resolved
Missing values (critical)	76 total	Minimal	Resolved
Customer_Segment variants	6	3	Resolved
Payment_Method variants	7	5	Resolved
Product_Category variants	10	5	Resolved
Country variants	10	5	Resolved
Customer_Name variants	18	12	Resolved
City variants	inconsistent	standardised	Resolved
Invalid Age values	1 (130+)	0	Resolved
Unit_Price data type	Text (Abc)	Float (#)	Resolved
Age_Group categories	—	4 categories	Created
Sales_Tier categories	—	4 categories	Created
Unit_Price_Scaled range	—	0 to 1	Created

Remaining nulls after cleaning: Email, Order_Date (original 5 missing + some unparseable formats), and Notes. These are acceptable as they are non-critical columns and removing those rows would lose valid order data.

8. Export

After validation, the cleaned dataset was exported from Tableau Prep by adding an Output step to the flow. The file was saved as a CSV for use in Tableau Desktop or further analysis. The final dataset contains 95 rows and 22 fields (18 original + 4 new calculated fields).

9. Pipeline Context: ETL vs ELT

Aspect	ETL	ELT
Where profiling happens	Before loading — in the transformation layer	After loading — inside the data warehouse
Tool used	Tableau Prep (this exercise)	SQL, dbt, Snowflake, BigQuery
Best for	Small datasets, strict governance	Big data, cloud warehouses, iterative transforms
Data quality control	Only clean data enters destination	Raw data lands first, cleaned in warehouse

Which approach is better for this dataset?

ETL is the correct approach for this exercise. The dataset is small (100 rows), the data quality issues are well-defined, and the destination is Tableau Desktop — not a cloud data warehouse. Tableau Prep is itself an ETL tool, so profiling and cleaning before loading is the natural fit. ELT would only be justified if this dataset were part of a larger pipeline feeding millions of rows into a warehouse like Snowflake or BigQuery.